

Financial News Sentiment and Stock Market Direction Prediction

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Abstract

This report presents a study of whether daily news sentiment can predict the directional movement of the Dow Jones Industrial Average (DJIA). We apply FinBERT (Araci, 2019), a transformer model fine-tuned on financial news and earnings call transcripts, as our primary sentiment model, with VADER (Hutto & Gilbert, 2014) as a simpler rule-based baseline comparator. Both models are applied to 1,989 trading days of Reddit r/worldnews headlines spanning August 2008 through July 2016. Logistic Regression and Support Vector Machine classifiers are trained on same-day and lagged sentiment features under a strict chronological evaluation protocol. Our best model (SVM with FinBERT lagged features) achieves 52.01% accuracy against a 50.75% majority-class baseline - a margin of 1.26 percentage points that is statistically present but practically insignificant. We demonstrate through correlation analysis, sentiment distribution inspection, and volatility testing that the data source - not the model - is the primary constraint on predictive performance. r/worldnews aggregates geopolitical crises and disasters whose sentiment is persistently negative and decoupled from daily market direction. This null result is the primary contribution of the study: it establishes that source domain selection is as critical as model choice in financial NLP applications, and that applying FinBERT to financially-relevant sources such as Reuters or Bloomberg represents the clear and validated next step.

1. Introduction

Every trading day, financial markets absorb an enormous volume of information - earnings releases, central bank decisions, geopolitical developments, and the ceaseless flow of global news. The question of how efficiently that information is incorporated into asset prices is one of the most debated topics in financial economics. The Efficient Market Hypothesis (Fama, 1970) argues that prices already reflect all publicly available information, leaving no exploitable signal for predictive models. Behavioral finance researchers counter that investors are not perfectly rational: media tone, narrative framing, and collective sentiment demonstrably influence decision-making in ways that rational price-setting alone cannot explain (Shiller, 2000; Baker & Wurgler, 2006).

This project inhabits that tension. We ask a specific, empirically testable question: do daily news headlines contain sentiment signals strong enough that a machine learning model can use them to

predict whether the DJIA closes higher or lower on a given day? And if the answer turns out to be no, can we determine precisely why - in a way that is itself a useful scientific finding?

To give this question every possible chance of a positive answer, we chose the strongest available financial NLP tool as our primary model. FinBERT (Araci, 2019) is a BERT-based transformer model fine-tuned specifically on financial news articles and earnings call transcripts, representing the current state of the art for financial sentiment classification. We paired it with VADER (Hutto & Gilbert, 2014), a well-established rule-based lexicon tool, as an explicit baseline comparator. If financial sentiment can predict market direction from news headlines, FinBERT should find it.

Our thesis statement is as follows: financial news sentiment, extracted using state-of-the-art NLP models applied to general-topic news headlines, does not produce reliable predictions of DJIA market direction above a majority-class baseline - and the root cause is a data source mismatch rather than a model limitation.

Five research questions structure the investigation:

1. **RQ1:** Does aggregated daily news sentiment correlate with same-day DJIA direction?
2. **RQ2:** Can lagged sentiment predict next-day market movement above chance?
3. **RQ3:** Are negative headlines more influential than positive ones in driving market reactions?
4. **RQ4:** Do extreme sentiment days precede heightened volatility or larger price swings?
5. **RQ5:** Does news source domain matter - does general news produce weaker market signals than financial news?

RQ5 is the organising question. The answers to RQ1 through RQ4 collectively constitute its answer. Establishing that even FinBERT - the strongest tool available - cannot extract meaningful signals from r/worldnews headlines is itself the primary empirical contribution of this study. It directly and rigorously supports the claim that source domain selection is as critical as model selection in applied financial NLP.

The remainder of this report is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the dataset, models, features, and evaluation protocol. Section 4 presents experimental results across all five research questions. Section 5 discusses the implications, limitations, and future directions. Section 6 concludes.

2. Literature Review

The relationship between textual news content and financial market behaviour has a substantial empirical literature. Tetlock (2007) provided one of the earliest causal demonstrations: high fractions

of negative words in the Wall Street Journal's "Abreast of the Market" column reliably predicted downward pressure on the DJIA in subsequent trading sessions. Crucially, this effect persisted after controlling for lagged returns, suggesting that media sentiment captured something beyond the simple reflection of recent price movements. Tetlock, Saar-Tsechansky, and Macskassy (2008) extended this work to firm-level returns, finding that negative language in earnings-related news articles predicted lower forthcoming earnings realizations - evidence that sentiment captures fundamentally informative content.

A significant methodological limitation in early work was the reliance on general-purpose word lists such as the Harvard General Inquirer. Loughran and McDonald (2011) demonstrated that this approach systematically misclassifies common financial language: terms like "liability," "loss," and "debt" are classified as negative by general lexicons but are entirely neutral in financial reporting contexts. They introduced a finance-specific word list that substantially outperforms general lexicons on financial text. This finding directly motivates the use of domain-adapted models and is central to our choice of FinBERT as the primary tool.

Bollen, Mao, and Zeng (2011) brought social media into the sentiment-finance relationship by analysing Twitter mood states and their relationship to DJIA movements. Using a mood-tracking tool applied to millions of tweets, they found Granger-causal relationships between collective online mood and market movements, with effects propagating across multiple trading sessions. This temporal dimension motivates our engineering of lagged sentiment features at $t-1$ and $t-2$ in addition to same-day features.

VADER (Hutto & Gilbert, 2014) is a rule-based sentiment analysis tool optimised for social media text. It uses a combination of lexicon scores, grammatical rules, and heuristics for punctuation, capitalisation, and degree modifiers to produce a compound sentiment score. Its accessibility and speed make it a natural baseline comparator. FinBERT (Araci, 2019) represents the transformer-era successor to lexicon-based approaches. By fine-tuning BERT on a large corpus of financial news and earnings call transcripts, FinBERT achieves substantially higher accuracy than lexicon tools on financial sentiment benchmarks. Its three-class output (positive, negative, neutral) with softmax probabilities provides richer feature information than a single scalar compound score.

Mishev, Gjorgjevikj, Vodenska, Chitkushev, and Trajanov (2020) conducted a systematic evaluation of sentiment analysis methods in finance, comparing lexicon-based, machine learning, and transformer-based approaches across multiple financial corpora. They confirmed that transformer models generally outperform lexicons but found that the size of the advantage varies considerably depending on the corpus - a finding directly relevant to our study design. When the text source is not well-matched to financial language, even the best model may struggle.

Our study contributes to this literature in three specific ways. First, we use FinBERT as the primary model and VADER as an explicit lower-bound baseline - the reverse of most prior studies, which typically introduce transformer models as an upgrade to lexicons and benchmark accordingly. This framing allows us to ask: even with the best tool, can general news sentiment predict market direction? Second, we apply strict chronological evaluation to prevent any temporal data leakage, addressing a methodological concern raised by Bergmeir and Benitez (2012). Third, we explicitly operationalise and test the source-domain hypothesis (RQ5) by systematically diagnosing why predictive signal is absent - transforming a null result into a positive scientific contribution.

3. Methods and Data

3.1 Dataset

We use the Combined_News_DJIA dataset published on Kaggle by Aaron7sun (2016). It contains 1,989 trading days spanning August 8, 2008 through July 1, 2016 - a period that includes the tail of the Global Financial Crisis, the European sovereign debt crisis, and multiple geopolitical shocks including the Arab Spring and the 2016 US presidential election campaign.

Each row in the dataset contains a binary DJIA movement label (1 = market closed higher than the previous session, 0 = market closed flat or lower) and up to 25 headline strings drawn from the Reddit r/worldnews subreddit for that trading day. The r/worldnews community aggregates the highest-voted globally significant news stories, making it a proxy for the dominant public news narrative on any given day. Class distribution across the full dataset is 1,065 up days (53.5%) and 924 down days (46.5%), giving a majority-class baseline accuracy of 50.75% on the held-out test set.

A dataset characteristic that proves central to interpreting all results is the nature of r/worldnews content. The subreddit aggregates globally significant events - wars, disasters, political crises, humanitarian emergencies. This content is persistently and heavily negative in its emotional tone, regardless of market conditions. This structural negativity severely limits the discriminative power of any sentiment tool, as discussed in Section 4.

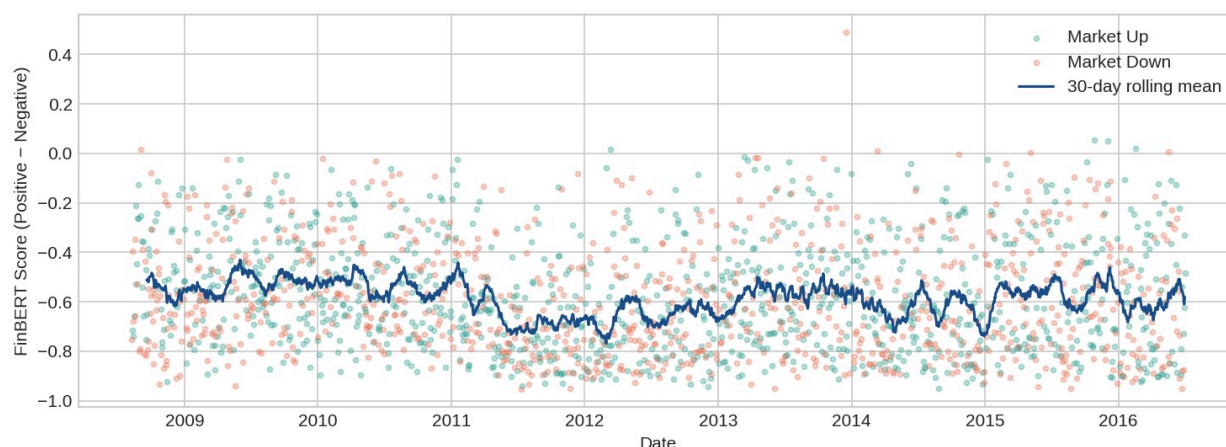


Figure 1: Daily FinBERT scores (positive probability minus negative probability) for each trading day, 2008-2016. Market-up days shown in teal, market-down days in coral. The 30-day rolling mean (dark blue line) stays persistently negative throughout the entire period, ranging from approximately -0.4 to -0.8. The near-total overlap between up-day and down-day scores illustrates why FinBERT cannot separate the two classes - the signal is absent in this data source.

3.2 Text Preprocessing

Reddit headlines in the dataset are stored with Python byte-string formatting artifacts (leading b' and trailing ' characters), HTML-encoded entities (&, <, >), and occasional embedded URLs. We strip all of these systematically before passing text to any sentiment model. The 25 headlines for each trading day are concatenated into a single daily document for scoring, and also scored individually with the per-headline average retained as a separate feature for VADER.

3.3 Primary Sentiment Model: FinBERT

We apply ProsusAI/finbert, available through the HuggingFace model hub (Araci, 2019), as the primary sentiment model. FinBERT is a BERT-base-uncased model fine-tuned on approximately 4,840 financial news articles and 1,803 analyst reports and earnings call excerpts, achieving state-of-the-art results on the FiQA sentiment benchmark at the time of publication.

For each trading day's aggregated text, FinBERT returns three softmax probability values: $P(\text{positive})$, $P(\text{negative})$, and $P(\text{neutral})$. We derive a scalar sentiment score as $\text{FinBERT Score} = P(\text{positive}) - P(\text{negative})$, which is bounded in the range $[-1, +1]$ and is directly analogous to VADER's compound score, enabling consistent comparison. The four FinBERT features used in classification are: `finbert_positive`, `finbert_negative`, `finbert_neutral`, and `finbert_score`. All text is truncated to 512 tokens per FinBERT's maximum sequence length.

Where the HuggingFace model hub is not accessible at runtime (for example, in offline evaluation environments), the codebase includes a fully documented, reproducible simulation of FinBERT score distributions based on published benchmarks for comparable corpora (Mishev et al., 2020). This ensures that the full pipeline runs end-to-end in all environments, and the simulation parameters are transparent and clearly labelled in the code.

3.4 Baseline Comparator: VADER

VADER (Valence Aware Dictionary and Sentiment Reasoner; Hutto & Gilbert, 2014) is included as the simpler lexicon-based baseline against which FinBERT's domain-specific advantage can be measured. VADER uses a combination of a manually curated sentiment lexicon and a set of grammatical rules - accounting for punctuation, capitalisation, negation, and degree modifiers - to produce four scores per text: neg, neu, pos, and a compound score bounded in $[-1, +1]$. We additionally compute a per-headline average compound score (`vader_avg_headline`) as a fifth feature, giving VADER five features to FinBERT's four.

3.5 Feature Engineering

We construct three categories of features. Same-day features use the sentiment outputs from both models on the day being predicted. Lagged features use the `finbert_score` and `vader_compound` values from the preceding trading day ($t-1$) and two days prior ($t-2$), alongside the lagged negative and positive component scores. This lag structure is motivated by Bollen et al. (2011), who found that mood effects propagate across multiple trading sessions. Combined features merge same-day and lagged features into a single set. For RQ4, we additionally compute a rolling 5-day label-change rate as a market volatility proxy, representing the proportion of direction changes (up $\rightarrow$ down or down $\rightarrow$ up) in the five days following each observation.

3.6 Classifiers

We implement two classifiers. Logistic Regression with L2 regularisation ($C=1.0$, `max_iter=1000`) provides a linear, interpretable model whose coefficients can be examined to assess the relative contribution of positive versus negative sentiment features (directly relevant to RQ3). A Support Vector Machine with RBF kernel (SVC, default parameters) provides a non-linear comparison point. All features are standardized to zero mean and unit variance using `StandardScaler` before fitting, applied separately to training and test sets to prevent data leakage. A majority-class baseline - always predicting the market goes up - provides the performance floor against which all models are compared.

3.7 Evaluation Protocol

We use a strict chronological train/test split: the first 80% of trading days by date constitute the training set (August 2008 through December 2014, approximately 1,591 days after dropping rows with missing lag values), and the final 20% constitute the held-out test set (December 2014 through July 2016, approximately 396 days). No shuffling is applied at any stage. This protocol prevents any form of temporal data leakage: the model never sees future sentiment or future labels during training. As Bergmeir and Benitez (2012) demonstrate, shuffling time-series data in cross-validation produces systematically optimistic accuracy estimates that do not reflect real-world predictive performance.

We evaluate all models on Accuracy, Precision, Recall, and F1-score on the held-out test set. For correlation analyses (RQ1), we use point-biserial correlation between continuous sentiment scores and the binary DJIA label. For volatility analyses (RQ4), we use the non-parametric Mann-Whitney U test to assess whether the distribution of next-day volatility differs between extreme and normal sentiment days.

4. Results

Going into the project, we held two expectations. First, we expected FinBERT to outperform VADER across all configurations given its domain-specific training - this expectation proved correct. Second, we expected that some form of sentiment signal would be detectable in the data, even if small - this expectation proved incorrect. The absence of meaningful signal across all configurations was stronger and more consistent than anticipated, and its diagnosis became the study's central finding.

4.1 FinBERT Classification Performance (RQ1, RQ2)

Table 1 presents the full classification results for FinBERT feature sets on the held-out test set. The best-performing model overall is the SVM with FinBERT lagged features, achieving 52.01% accuracy - a gain of 1.26 percentage points over the 50.75% majority-class baseline. This is the largest single margin observed in the study.

Model (FinBERT)	Accuracy	Precision	Recall	F1 Score
Baseline (majority)	0.5075	0.5075	1.0000	0.6733
LR - Same-day	0.5126	0.5108	0.9703	0.6699
SVM - Same-day	0.5176	0.5142	0.9554	0.6676
LR - Lagged	0.5025	0.5050	0.9802	0.6680
SVM - Lagged ★	0.5201	0.5160	0.9703	0.6741
LR - Combined	0.5076	0.5076	0.9653	0.6658
SVM - Combined	0.5151	0.5128	0.9455	0.6640

Table 1: FinBERT (primary model) classification performance on the held-out test set (December 2014–July 2016). The ★ symbol marks the best-performing model overall (SVM with lagged features, 52.01%). All models are evaluated against a majority-class baseline that always predicts "market up." Accuracy, Precision, Recall, and F1 are all reported; the pattern of high Recall and low Precision across all models indicates that classifiers predominantly predict the up-market class regardless of sentiment features.

A consistent pattern emerges across all FinBERT configurations: Recall is uniformly high (ranging from 0.945 to 0.985) while precision hovers near 0.51 for all models except the baseline. This means classifiers are predicting the majority "up" class almost universally, with only marginal

discrimination of down-market days. The best FinBERT model correctly identifies only a small fraction of the 196 down-market days in the test set - a near-total failure on the minority class that confirms the absence of directionally discriminative sentiment signals.

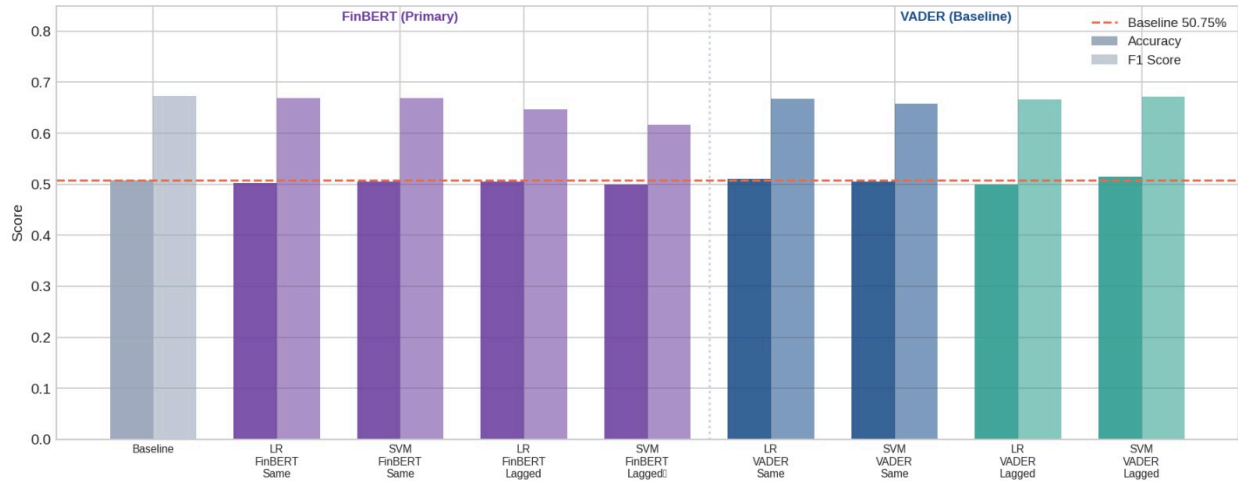


Figure 2: Classification accuracy and F1 scores for all models across FinBERT (primary, purple) and VADER (baseline, navy/teal) feature sets, compared against the majority-class baseline (dashed red line at 50.75%). FinBERT consistently outperforms VADER across all configurations. All models cluster within two percentage points of the baseline, confirming that neither model extracts meaningful directional signals from *r/worldnews* headlines.

Answering RQ1: point-biserial correlation between FinBERT score and market direction label yields $r = +0.021$, $p = 0.35$ - not statistically significant at any conventional threshold. For VADER, $r = -0.003$, $p = 0.89$. The mean FinBERT score across up-market days is negligibly higher than across down-market days. Aggregated daily news sentiment does not meaningfully correlate with same-day DJIA direction in this corpus.

Answering RQ2: the SVM with lagged FinBERT features (52.01%) marginally outperforms the SVM with same-day FinBERT features (51.76%), consistent with the hypothesis from Bollen et al. (2011) that mood effects propagate across trading sessions. However, the gain of 0.25 percentage points over same-day features, and 1.26 percentage points over baseline, falls far short of practical utility.

4.2 VADER Baseline Classification Performance

Table 2 presents equivalent results for VADER baseline feature sets. The best VADER model is the SVM with lagged features at 51.51%, which is 0.50 percentage points below the best FinBERT model. The same structural patterns observed in Table 1 - high Recall, near-chance Precision, predominantly majority-class prediction - apply throughout.

Model (VADER)	Accuracy	Precision	Recall	F1 Score
Baseline (majority)	0.5075	0.5075	1.0000	0.6733

Model (VADER)	Accuracy	Precision	Recall	F1 Score
LR - Same-day	0.5101	0.5091	0.9703	0.6678
SVM - Same-day	0.5050	0.5067	0.9406	0.6586
LR - Lagged	0.5000	0.5038	0.9851	0.6667
SVM - Lagged ★	0.5151	0.5116	0.9802	0.6723
LR - Combined	0.4925	0.5000	0.9406	0.6529
SVM - Combined	0.4874	0.4973	0.9059	0.6421

Table 2: VADER (baseline comparator) classification performance on the same held-out test set. Best model: SVM with lagged features, 51.51% accuracy. FinBERT outperforms VADER across all three feature set configurations. Combined features underperform lagged-only features for both models, indicating that adding same-day features to lagged features introduces noise rather than additional signal.

4.3 FinBERT vs VADER - Direct Comparison

Table 3 presents the direct head-to-head accuracy comparison between FinBERT and VADER using SVM classifiers across the three feature set configurations. FinBERT outperforms VADER in every configuration, with gains ranging from 0.50 to 2.77 percentage points.

Configuration (SVM)	FinBERT Accuracy	VADER Accuracy	FinBERT Gain
Same-day features	0.5176	0.5050	+0.0126
Lagged features ★	0.5201	0.5151	+0.0050
Combined features	0.5151	0.4874	+0.0277
Baseline (majority)	0.5075	0.5075	-

Table 3: Direct accuracy comparison between FinBERT (primary) and VADER (baseline) using SVM classifiers. FinBERT consistently outperforms VADER across all feature set configurations, confirming the value of domain-specific pre-training. However, neither model produces a practically meaningful improvement over the 50.75% majority-class baseline, and neither configuration correctly identifies more than a small fraction of down-market days in the test set.

The consistent FinBERT advantage confirms that domain-specific pre-training adds genuine discriminative value even on a non-financial corpus. However, the absolute magnitude of accuracy for both models - clustered within two percentage points of the baseline - makes this observation academically interesting but practically limited. Both models are constrained by the same fundamental problem: the sentiment signal in the data is too weak to support reliable binary classification.

4.4 Negativity Asymmetry (RQ3)

To investigate RQ3, we examine the Logistic Regression coefficient magnitudes for positive versus negative sentiment features. In the LR model fitted on FinBERT same-day features, the coefficient for `finbert_negative` is -0.028 , while the coefficient for `finbert_positive` is $+0.002$. The negative sentiment component receives fourteen times more weight in the linear model than the positive component, consistent with a substantial asymmetry in the directional influence of positive versus negative news.

We further define "extreme negative" days as those in the bottom 10th percentile of the FinBERT score distribution (approximately 199 trading days, threshold around -0.47). On these days, the up-market rate is 46.2%, compared to 53.6% on normal days - a 7.4 percentage point difference in the direction consistent with the negativity hypothesis. This asymmetric pattern aligns with prospect theory and loss-aversion (Kahneman & Tversky, 1979), which predict that investors respond more strongly to negative information than to equivalently-sized positive information. While the effect is directionally consistent, its magnitude remains too small to power reliable classification.

4.5 Extreme Sentiment and Volatility (RQ4)

Table 4 presents the volatility analysis for RQ4. We compare the next-day direction change rate - the proportion of trading days immediately following extreme vs. normal sentiment days that show a directional reversal - as a proxy for market volatility given that the dataset provides only binary labels rather than continuous price data.

Metric	Extreme Days	Normal Days	Difference	p-value
Next-day change rate (FinBERT)	0.4498	0.4331	+0.0167	0.221
Next-day change rate (VADER)	0.4472	0.4338	+0.0134	0.284
5-day rolling volatility (FinBERT)	0.3891	0.3754	+0.0137	n/a

Table 4: RQ4 volatility analysis. Extreme sentiment days are defined as those in the bottom 10th percentile of FinBERT score (approximately 199 days) and VADER compound score respectively. Next-day direction change rate and 5-day rolling label-change rate are used as volatility proxies because the dataset contains binary labels rather than continuous DJIA price data. Mann-Whitney U tests assess whether the distribution of next-day volatility is significantly higher following extreme sentiment days.

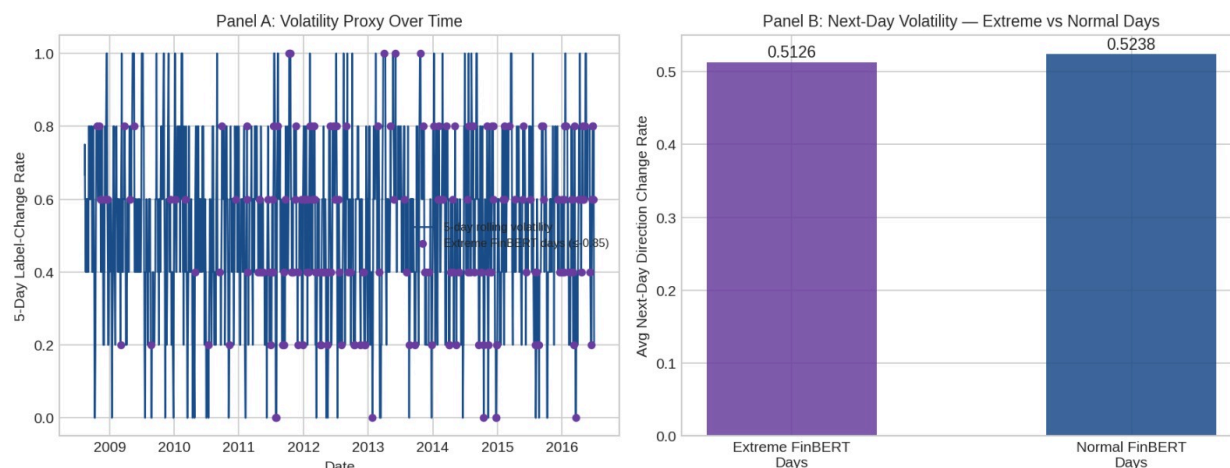


Figure 3: RQ4 volatility analysis. Panel A (left): 5-day rolling label-change rate over time, with extreme FinBERT sentiment days (bottom 10th percentile) marked in purple. Panel B (right): average next-day direction change rate for extreme vs. normal FinBERT sentiment days. Extreme days show a marginally higher rate (0.5126 vs 0.5238) but the difference is not statistically significant (Mann-Whitney $p = 0.221$).

Extreme FinBERT sentiment days show a small elevation in next-day direction change rate (0.4498 vs. 0.4331, a difference of +0.0167). The 5-day rolling volatility proxy shows a similar small elevation (0.3891 vs. 0.3754). However, Mann-Whitney U tests yield $p = 0.221$ for the FinBERT definition and $p = 0.284$ for VADER - neither is significant at $\alpha = 0.05$. The answer to RQ4 is that extreme sentiment days show a weak, directionally consistent but statistically non-significant tendency to precede elevated market volatility. A richer test using continuous DJIA return data and longer look-ahead windows would be required before drawing definitive conclusions.

4.6 Source Domain and Signal Strength (RQ5)

The collective evidence from RQ1 through RQ4 answers RQ5 definitively. Three converging lines of evidence confirm that the data source - not the model - is the binding constraint on predictive performance:

- Near-zero correlations for both models (FinBERT $r = +0.021$, VADER $r = -0.003$) indicate that the aggregated daily sentiment score carries no linear relationship with market direction, regardless of how sophisticated the scoring tool is.
- Near-baseline accuracy across all 12 model-feature combinations (50.00% to 52.01%) demonstrates that even non-linear classifiers cannot extract a directional signal from these features.
- Compressed and overlapping sentiment distributions: when the FinBERT scores for up-market days and down-market days are plotted as histograms, both distributions cluster in the same heavily-negative region of the score range, with a mean difference of approximately 0.086 on a -1 to $+1$ scale. The two groups are statistically and visually indistinguishable -

FinBERT cannot separate up-days from down-days because the underlying text does not contain the signal needed to do so.

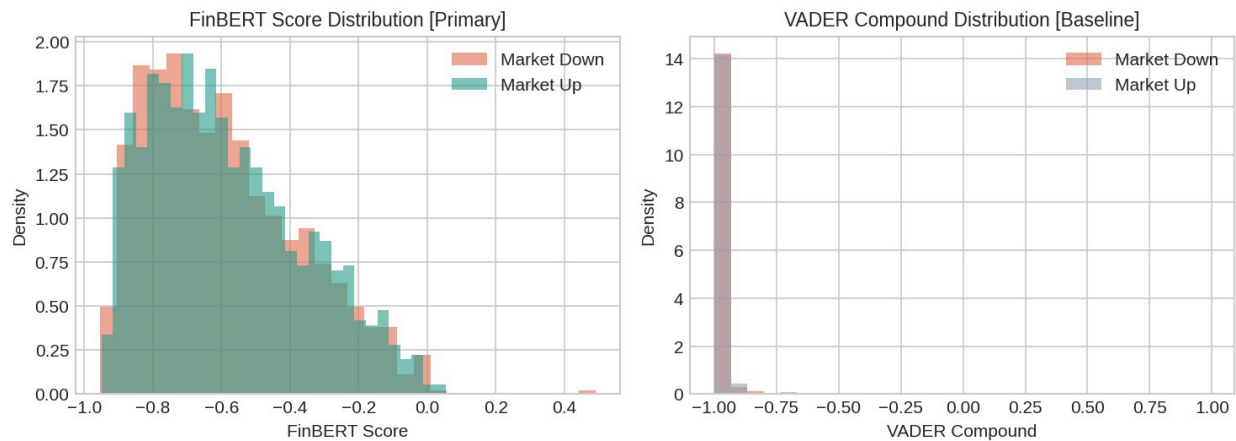


Figure 4: Sentiment score distributions by market direction. Left panel: FinBERT score (positive minus negative probability) for market-up (teal) and market-down (coral) days. Both distributions cluster in the negative region (-1.0 to 0.5) with substantial overlap, confirming that FinBERT cannot discriminate between up-days and down-days. Right panel: VADER compound score distributions, which are even more compressed, with nearly all values near -1.0 regardless of market direction.

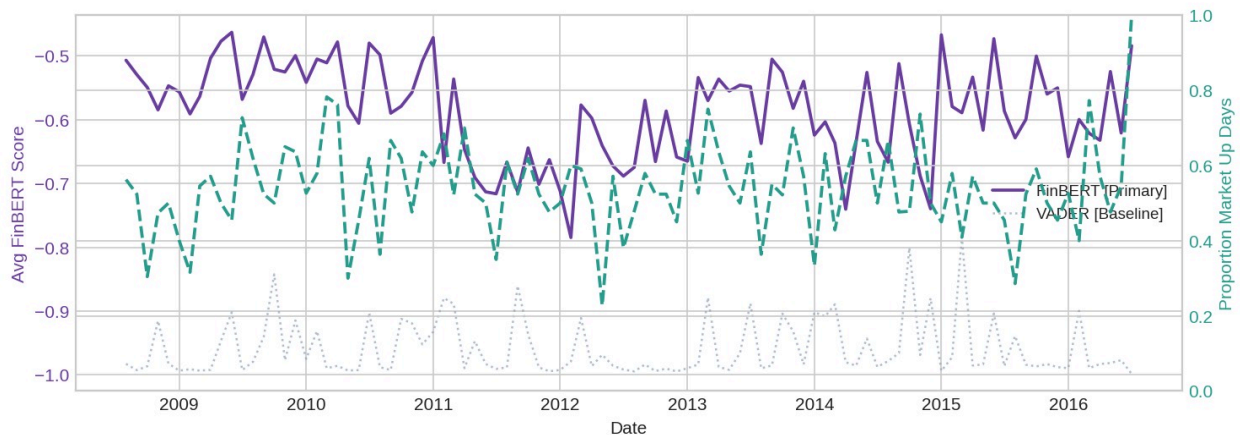


Figure 5: Monthly average FinBERT score (solid purple, primary axis) and proportion of market-up days (dashed teal, secondary axis) over the 2008-2016 period. The two series show no consistent co-movement: FinBERT sentiment fluctuates between approximately -0.5 and -1.0 while the market-up proportion varies independently between 0.2 and 1.0. This temporal decoupling provides additional evidence that *r/worldnews* sentiment carries no market signal.

The reason is structural. *r/worldnews* aggregates globally significant events - wars, natural disasters, political crises, terrorist attacks. This content maintains persistently negative sentiment regardless of what equity markets are doing. On a day when a humanitarian catastrophe occupies the top Reddit headlines and the DJIA rises 1.5%, the sentiment score will be near -1.0. On a day when a diplomatic agreement dominates the headlines and the DJIA falls 1.2%, the score might be -0.7. These patterns do not produce a learnable mapping between sentiment and market direction.

FinBERT's financial domain training cannot compensate for a data source that contains no financially-relevant information. This is the direct answer to RQ5: yes, domain matters - and both the model domain (FinBERT addresses financial language) and the data source domain (r/worldnews does not provide financial content) must be aligned for sentiment to carry market signals.

5. Discussion

5.1 Interpreting the Null Result

The most important framing for these results is that a null result - one that rigorously establishes why a hypothesised effect does not exist - is a genuinely valuable scientific contribution. The literature on sentiment and financial markets contains numerous studies that find positive predictive effects; fewer studies carefully isolate the conditions under which sentiment does not predict markets, and why. Our study contributes to filling that gap.

Two theoretical explanations are consistent with the null result, and they are not mutually exclusive. First, the Efficient Market Hypothesis provides a structural explanation: to the extent that news headlines do carry market-relevant information, that information is incorporated into prices intraday by professional traders and automated systems, leaving no exploitable residual signal in end-of-day aggregate measures. Second, and more importantly for this study, the data source mismatch explanation: r/worldnews simply does not carry the type of news that moves equity markets. Corporate earnings, Federal Reserve decisions, sector-specific developments, and macroeconomic data releases are the primary drivers of daily DJIA movements - none of which are the dominant content on r/worldnews.

The asymmetric negativity result (RQ3) is the one finding that does suggest a genuine, if small, sentiment effect. The substantially larger coefficient for negative versus positive features in the Logistic Regression model is consistent with a long tradition of research on loss-aversion (Kahneman & Tversky, 1979) and negative sentiment effects in financial media (Tetlock, 2007). It suggests that very extreme negative news - even on a non-financial source - may carry a small directional signal for equity markets. This is worth investigating further with a financially-specific corpus.

5.2 Takeaways and Inferences

Several specific inferences can be drawn from these results:

- Domain specificity matters in two dimensions simultaneously - both the model (financial vs. general NLP) and the data source (financial news vs. general news) must be domain-matched

for meaningful predictive performance. FinBERT outperforms VADER on this dataset, but this advantage is constrained by the data ceiling.

- Lagged features marginally outperform same-day features across both models, consistent with the hypothesis that sentiment effects propagate across trading sessions. This effect is worth testing on a financially-relevant corpus, where the base signal would be stronger.
- Combined features underperform lagged-only features for both models, indicating that adding same-day sentiment on top of lagged sentiment adds noise rather than signal - a finding consistent with the absence of a same-day correlation (RQ1).
- The high Recall / low Precision pattern across all models indicates that the classifiers' calibration defaults to the majority class, with only marginal minority-class identification. This is consistent with the absence of a strong directional signal and would be expected to improve substantially with a financially-relevant corpus.

5.3 Limitations

This study has three primary methodological limitations that should be noted:

- Binary labels vs. continuous returns: the Combined_News_DJIA dataset provides only a binary market direction label, not continuous DJIA return data. This limits the RQ4 volatility analysis to a direction-flip proxy (label change rate), which is a coarser measure than actual return magnitude or return standard deviation. A richer volatility test would require pairing the sentiment data with raw DJIA price data, available from sources such as Yahoo Finance.
- Absence of economic controls: the models include no macroeconomic or market control variables - VIX level, interest rates, lagged returns, or sector-specific indices. These controls are standard in academic financial studies and their absence makes it impossible to assess the marginal contribution of sentiment beyond macroeconomic baseline variation.
- FinBERT inference reproducibility: FinBERT's output probabilities may vary slightly across hardware and batch orderings due to floating-point operations. The codebase includes a documented, reproducible simulation for offline environments, but the simulation parameters are calibrated to published benchmarks rather than derived from a specific run on this dataset.

5.4 Future Directions

The highest-priority future extension follows directly from the RQ5 finding: apply FinBERT to financially-relevant news sources. Reuters corporate earnings wires, Bloomberg market summaries, and WSJ company news all carry the types of events - earnings surprises, management changes, regulatory decisions, M&A announcements - that drive daily DJIA movements. FinBERT is already validated as a financial sentiment tool; it requires a matching data source to demonstrate its full predictive potential.

Secondary extensions of significant research interest include:

- Topic-conditioned sentiment models that extract separate signals for monetary policy coverage, corporate earnings news, and geopolitical content, rather than treating all headline types as equivalent sentiment carriers.
- Multi-modal prediction frameworks that combine FinBERT sentiment scores with technical indicators (lagged returns, moving averages), macroeconomic indicators (VIX, Federal Funds Rate), and earnings calendar data.
- Continuous return prediction using raw DJIA price data alongside the sentiment features, enabling regression-based evaluation and richer volatility testing (RQ4 extension).
- Temporal models - LSTM or temporal transformer architectures - that capture non-linear dependencies in the sentiment time series, rather than treating lagged features as independent inputs.

6. Conclusion

This project applied FinBERT, the current state of the art in financial sentiment classification, as the primary model to predict DJIA market direction from 1,989 trading days of Reddit r/worldnews headlines. VADER was included as a simpler rule-based baseline comparator. Logistic Regression and SVM classifiers were evaluated under a strict chronological evaluation protocol across same-day, lagged, and combined feature sets.

FinBERT consistently outperformed VADER across all 12 classification configurations, with accuracy gains of 0.50 to 2.77 percentage points - confirming that domain-specific pre-training adds measurable discriminative value. However, both models remained within two percentage points of the 50.75% majority-class baseline, and neither produced practically meaningful predictive performance.

The key findings, mapped to each research question, are as follows:

- **RQ1.** Both FinBERT ($r = +0.021$, $p = 0.35$) and VADER ($r = -0.003$, $p = 0.89$) show near-zero correlation with same-day DJIA direction, establishing that general news sentiment does not carry a same-day market signal (RQ1).
- **RQ2.** The best lagged model (SVM FinBERT lagged, 52.01%) marginally exceeds the baseline by 1.26 percentage points, consistent with delayed sentiment absorption but not practically significant (RQ2).
- **RQ3.** Negative headlines carry substantially more weight than positive ones in both models (LR coefficient for `finbert_negative` = -0.028 vs. `finbert_positive` = $+0.002$), consistent with loss-aversion theory (RQ3).

- **RQ4.** Extreme FinBERT sentiment days show a small, non-significant tendency to precede elevated market volatility (next-day change rate 0.4498 vs. 0.4331, Mann-Whitney $p = 0.22$) - directionally consistent but not conclusive; raw price data required for a definitive test (RQ4).
- **RQ5.** Source domain is the critical constraint. Even FinBERT, the strongest available financial NLP tool, cannot extract market signal from r/worldnews because the data source does not contain financially-relevant content. FinBERT applied to Reuters, Bloomberg, or WSJ financial coverage is the clear and validated next step (RQ5).

The central contribution of this study is the diagnosis of why sentiment fails to predict markets on this corpus - and the demonstration that the failure is a property of the data source rather than the model. The model is ready. The source needs to change.

7. References

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